

Thermal hotspot detection in solar panel using AI and infrared imaging

A Project Report

Abstract

Undetected faults in solar photovoltaic panels — cracked cells, failing bypass diodes, loose connections, and localized shading — often manifest first as small, invisible-to-the-eye rises in surface temperature long before they show up as a measurable drop in energy output. Conventional thermal inspection relies on handheld infrared cameras or drone-based surveys carried out at fixed intervals, which means a fault can sit undetected for weeks between inspections. This project presents a low-cost, self-contained prototype built around a Raspberry Pi 4 that combines a low-resolution thermal sensor (MLX90640) with a standard visual camera to continuously monitor a panel for abnormal heating. The captured thermal data is interpolated, registered against the visual frame, and passed through a YOLOv8-nano object detection model trained to recognise and localize hotspot regions. Electrical readings from an INA219 voltage/current sensor are logged alongside the thermal data so that a detected hotspot can be checked against an actual dip in power output, giving the system a way to distinguish a real fault from ordinary heating. The result is a portable diagnostic unit that brings a basic form of automated thermal inspection within reach of small installations that would not otherwise have access to industrial-grade thermography equipment.

Chapter 1: Introduction

1.1 Background

The push toward renewable energy has made solar photovoltaic (PV) installations one of the fastest-growing sources of electricity generation worldwide. As these installations scale from a handful of rooftop panels to large arrays, the question of how to keep every panel performing at its rated efficiency becomes harder to answer manually. A single damaged cell can drag down the output of an entire string of panels, and because PV modules are usually mounted in places that are inconvenient to inspect closely and often not wired with per-panel monitoring, a fault can go unnoticed until it shows up as an unexplained dip in the monthly generation report.

Thermal imaging is the established way to catch these faults early, because a damaged or shaded cell tends to run hotter than its neighbours — a phenomenon well documented in PV fault literature. What has kept thermal inspection from becoming a routine, continuous practice is cost and manual effort: professional thermal cameras are expensive, and interpreting a thermal image still usually requires a trained eye. This project explores whether a cheap sensor, a small computer, and a lightweight machine learning model can be combined to automate that interpretation step, at least for small-scale deployments.

1.2 Motivation

The motivation for this project came from a simple observation: most of the tools that make thermal fault detection reliable — high-resolution thermal cameras, drone platforms, trained inspection software — are built for utility-scale solar farms and priced accordingly. A homeowner with a rooftop array, or a college running a small solar demonstration project, has no equivalent option. Building a prototype around commodity components (a Raspberry Pi, a \$50-class thermal sensor, an ordinary Pi camera) was a way to test how far a low-cost setup could go toward the same goal.

1.3 Objectives

- Design and assemble a portable hardware unit capable of capturing synchronized thermal and visual images of a PV panel.
- Develop a Python-based pipeline to acquire raw thermal data from the MLX90640 and interpolate it into a usable resolution.
- Register the thermal and visual image frames so a detected anomaly can be mapped to its exact physical location on the panel.
- Train and deploy a YOLOv8-nano model for automatic hotspot detection and localization on-device.
- Log voltage and current via the INA219 sensor to correlate thermal anomalies with actual electrical performance loss.
- Present results on an integrated touchscreen display for immediate feedback in the field.

1.4 Scope and Limitations

This work is scoped as a functional prototype rather than a field-ready commercial product. It is intended for single-panel or small-array inspection — handheld scanning, or a fixed unit mounted to periodically check a small rooftop system — rather than continuous monitoring of a utility-scale solar farm. The MLX90640's native resolution (32x24) is modest, so the interpolation and

detection stages are designed to compensate for coarse input rather than assuming high-fidelity thermal imagery. Cloud connectivity, historical dashboards, and multi-panel fleet management are identified as extensions beyond the current scope, discussed further in Chapter 7.

Chapter 2: Literature Review

Thermal imaging as a diagnostic tool for photovoltaic systems has been studied extensively, with most existing work falling into two categories. The first uses high-resolution, industrial-grade infrared cameras mounted on drones to survey large solar farms, producing detailed thermal maps that are later reviewed by inspection software or trained personnel. These systems are accurate but expensive, and they are typically deployed on a scheduled basis rather than continuously. The second category focuses on fixed, panel-level monitoring using cheaper thermal sensors, often paired with basic threshold-based anomaly detection rather than trained machine learning models.

More recent research has started applying convolutional neural networks and object detection architectures to thermal images of solar panels, treating hotspot identification as a standard object detection problem. YOLO-family models in particular have been favoured in embedded and edge-deployment contexts because of their relatively low computational footprint compared to two-stage detectors, making them a practical fit for single-board computers like the Raspberry Pi. This project draws on that direction, applying the smallest variant in the YOLOv8 family specifically because it needs to run inference on a Pi 4 without a dedicated GPU or accelerator.

A gap that recurs across the literature is the reliance on a single imaging modality — either thermal alone or a professionally aligned thermal-visual pair captured with matched, calibrated cameras. Consumer-grade thermal sensors like the MLX90640, when paired with an independent visual camera, need their own registration step to align two frames captured from physically different sensor positions, which is a practical detail that lower-cost prototypes have to solve themselves rather than inheriting from expensive, pre-calibrated dual-sensor hardware. This project treats that registration step as a distinct stage in the pipeline for exactly that reason.

Chapter 3: Problem Statement and Objectives

3.1 Problem Statement

Existing methods for detecting thermal faults in solar panels are either too expensive for small-scale deployment (professional thermal cameras, drone surveys) or too infrequent to catch

developing faults early (scheduled manual inspection). There is a need for a low-cost, automated system capable of continuously or periodically screening PV panels for abnormal heating and flagging the exact location of a potential fault without requiring a trained technician to interpret the thermal image.

3.2 Objectives (Restated)

The objectives outlined in Section 1.3 guide the system design described in the following chapters: building the hardware unit, implementing the four-stage software pipeline, training the detection model, and validating results against electrical performance data.

Chapter 4: System Design and Methodology

4.1 System Overview

The system is built around a Raspberry Pi 4 acting as the central processing unit, connected to a thermal sensor and a visual camera for image capture, an INA219 and DS18B20 for electrical and reference temperature data, and an HDMI touchscreen for on-site display. The overall data flow proceeds through four stages, summarized in Table 4.1.

Stage	Function
1. Data Acquisition & Interpolation (Python)	Reads raw MLX90640 frames over I2C and upsamples the 32x24 grid into a smoother thermal map; captures matching RGB frame from Pi Camera
2. Preprocessing & Thermal-Visual Registration	Aligns thermal and visual frames geometrically; normalizes temperature values for consistent model input
3. Computer Vision Hotspot Localization	Scans the registered thermal map for regions with abnormal temperature relative to surrounding cells
4. YOLOv8-nano Detection	Lightweight object detection model classifies and draws bounding boxes around confirmed hotspot regions, optimized to run on Pi 4 without a GPU

Table 4.1: Software pipeline stages

4.2 Data Acquisition and Interpolation

The MLX90640 communicates with the Raspberry Pi over I2C and outputs a 32x24 grid of temperature readings per frame. Because this resolution is too coarse to localize a hotspot precisely, each frame is upsampled using interpolation (for example, bicubic interpolation) to

produce a smoother thermal map before any further processing. The Pi Camera Module 3 captures a corresponding 12MP visual frame of the same panel at roughly the same moment, giving the system a matched pair of thermal and visual images for every capture cycle.

4.3 Preprocessing and Thermal-Visual Registration

Because the thermal and visual cameras are mounted separately and have different fields of view, a coordinate in the thermal frame does not automatically correspond to the same coordinate in the visual frame. This stage applies a geometric transformation to align the two, so that once a hotspot is found in the thermal data, its location can be marked accurately on the visual image as well. Temperature values are also normalized at this stage to keep the input range consistent for the detection model, regardless of ambient conditions on a given day.

4.4 Computer Vision Hotspot Localization

With the two frames registered, the system scans the thermal map for regions whose temperature deviates significantly from the surrounding area of the panel. This step narrows down candidate regions before they are passed to the detection model, reducing false positives that might otherwise arise from uniform heating caused simply by direct sunlight.

4.5 Detection with YOLOv8-nano

YOLOv8-nano, the smallest model in the YOLOv8 family, performs the final classification and bounding-box localization of hotspot regions. It was chosen specifically for its small parameter count, which allows it to run inference directly on the Raspberry Pi 4's CPU within a reasonable time frame, avoiding the need for an external GPU or cloud-based inference. The model is trained on a labelled dataset of thermal images showing both normal and faulty panel regions, so that it learns to distinguish an actual defect from benign thermal variation.

4.6 Electrical Correlation

In parallel with imaging, the INA219 continuously logs the panel's voltage and current output, and the DS18B20 records a reference contact/ambient temperature. When the vision pipeline flags a hotspot, the system checks the electrical log around that time window to see whether the flagged region coincides with a measurable drop in power output. This step is what allows the system to move beyond simply noticing a warm patch and toward indicating whether that warm patch is likely to be an actual performance-affecting fault.

Chapter 5: Hardware Implementation

Table 5.1 lists the complete bill of materials used in the prototype, along with the role each component plays in the system.

Component	Specification	Purpose
Thermal Sensor	MLX90640 (32x24 IR array)	Captures raw thermal image of panel surface
Processing Unit	Raspberry Pi 4 (4GB/8GB)	Runs acquisition, preprocessing, and inference
Visual Camera	Pi Camera Module 3 (12MP)	Captures reference visual image
Display	3.5"/5" HDMI Touchscreen	On-device result display
Storage	32GB microSD (Class 10)	OS, model weights, and logged data
Power	USB-C Power Bank	Field portability
Cabling	Short HDMI-to-micro-HDMI	Connects display to Pi
Current/Voltage Sensor	INA219	Correlates hotspots with electrical output
Temperature Sensor	DS18B20	Ambient/reference temperature logging

Table 5.1: Bill of Materials

The Raspberry Pi 4 sits at the centre of the assembly, with the MLX90640 and DS18B20 connected over I2C/one-wire, the Pi Camera Module 3 connected via the CSI ribbon interface, and the INA219 also on the I2C bus, sharing it with the thermal sensor through distinct addresses. The touchscreen connects over HDMI (via the short HDMI-to-micro-HDMI cable) with touch input handled over USB, giving the unit a fully self-contained display without needing an external monitor. A USB-C power bank supplies power to the Pi, making the whole assembly field-portable for handheld or tripod-mounted scanning of a panel.

Chapter 6: Results and Discussion

As this report documents the design and build phase of the project, the results discussed here are the expected behaviour of the system based on the pipeline design, pending full field testing. A successfully operating unit should be able to capture a thermal-visual frame pair, process it through interpolation and registration, and return a bounding box around any detected hotspot

within a few seconds — fast enough for practical handheld or spot-check use. Where the electrical correlation stage is active, the system should additionally indicate whether a flagged hotspot corresponds to a measurable dip in the panel's voltage or current output, which would distinguish it from a false alarm caused by uneven sunlight or reflection.

The main constraints observed during development relate to the resolution of the MLX90640: because its native output is only 32x24 pixels, the quality of the final detection is heavily dependent on how well the interpolation and registration stages compensate for that limitation. Detection accuracy on very small or early-stage faults is expected to be lower than what a professional high-resolution thermal camera would achieve, which is a reasonable trade-off given the target of keeping the overall system low-cost.

Chapter 7: Conclusion and Future Scope

7.1 Conclusion

This project set out to build a low-cost, portable alternative to expensive thermal inspection equipment for solar panels, using commodity hardware and a lightweight AI detection pipeline. By combining a low-resolution thermal sensor with a standard camera, aligning the two, and running a compact YOLOv8-nano model on a Raspberry Pi 4, the system demonstrates that a meaningful degree of automated fault detection is achievable without industrial-grade equipment or cloud infrastructure. Correlating detected hotspots with actual electrical readings adds a layer of validation that helps separate genuine faults from ordinary thermal variation, which is a detail that purely image-based systems often skip.

7.2 Future Scope

- Extending the system to monitor multiple panels simultaneously, either through a networked array of units or a moving/scanning single unit.
- Adding cloud connectivity and a historical dashboard so degradation trends can be tracked over time rather than viewed only per-scan.
- Training the YOLOv8-nano model on a larger, more diverse dataset of real fault cases to improve detection accuracy.
- Exploring a higher-resolution thermal sensor as a drop-in upgrade once cost allows, to test how much the current bottleneck is resolution versus model accuracy.
- Adding automated alerting (SMS/email/app notification) when a hotspot correlated with power loss is detected.

References

[1] Literature and datasheets referenced for the MLX90640 thermal sensor, Raspberry Pi 4 hardware documentation, and the Ultralytics YOLOv8 model family should be listed here in the citation style required by your department (IEEE/APA).

[2] Add specific papers on PV thermal fault detection and YOLO-based defect detection that were consulted during the literature review, once finalized.